PROJECT REPORT

PKM Urdu Assistant

An AI-powered Urdu question-answering system for Punjab Police Khidmat Markaz services

پنجاب پولیس خدمت مرکز — اردو معلوماتی اسسٹنٹ

Independent project. Not an official Punjab Police product. Service information was collected from the public website pkm.punjab.gov.pk.

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Abstract

Citizens of Punjab who need services from Punjab Police Khidmat Markaz (PKM) often struggle to find accurate information about fees, required documents, processing times and centre locations. The information is published on the PKM website but is spread across dozens of pages, and most people prefer to ask in Urdu, frequently typed in Roman script.

This project developed the **PKM Urdu Assistant**, a conversational system that answers such questions in Urdu. Information was collected from the official website with automated scrapers, cleaned and structured into a knowledge base covering 14 services and 93 service centres in all 36 districts. The assistant uses a multilingual sentence-embedding model to find the most relevant facts and, through Retrieval-Augmented Generation (RAG), uses Google's Gemini model to phrase natural Urdu answers strictly from those facts. Rule-based components detect districts, service names and the aspect being asked about, so that a question about a fee returns only the fee.

The system is delivered as a web application, a desktop application and a command-line tool, and can be deployed free of cost on Hugging Face Spaces. It handles Urdu spelling variations, Roman Urdu and follow-up questions, refuses questions outside its knowledge, and continues to work in a basic mode when the language model is unavailable. On a labelled evaluation set, the assistant answered 86.7% of questions correctly in the first run (26/30); after error analysis and targeted fixes it answered all 33 questions of the extended set correctly.

Keywords: Urdu NLP, chatbot, Retrieval-Augmented Generation, sentence embeddings, e-governance, Punjab Police Khidmat Markaz, Gemini, Gradio

14	36	93	111	3
services	districts	service centres	knowledge entries	interfaces

1 Introduction

Punjab Police established Khidmat Markaz (service centres) across all 36 districts of Punjab to provide citizen services through a one-window operation, with the Punjab Information Technology Board (PITB) as technology partner.

Through these centres citizens obtain character certificates and police verification, renew or replace driving licences, register tenants and employees, verify vehicles, report lost items and crimes, and obtain copies of FIRs. Under the integrated system, any service can be obtained from the centre of any district.

Before visiting, a citizen needs to know which documents to bring, what the fee is, how long the service takes and where the nearest centre is. This information is published on the PKM website, but finding it requires navigating many tabs and pages, and the website does not answer questions directly.

Conversational assistants (chatbots) have become a common way to make such information accessible. However, most are built for English and depend on exact keywords. Urdu brings additional difficulties: it is written in a right-to-left script with many spelling variations, and a large share of users type it in Roman script.

The **PKM Urdu Assistant** was built to close this gap. It allows a citizen to ask a question in everyday Urdu, such as کریکٹر سرٹیفیکیٹ کی فیس کتنی ہے؟ (“What is the fee for a character certificate?”), and receive a short, accurate answer drawn from the official information.

2 Problem Statement

Citizens seeking PKM services face the following problems:

1. **Fragmented information.** Each of the 14 services has its own page with separate sections for documents, fees, timing and delivery, and centre details are split across 36 district tabs. Answering a simple question requires several clicks and careful reading.
2. **Incomplete preparation and repeat visits.** Citizens frequently arrive without a required document, such as an attested copy, an affidavit or a photograph of a specific size, and must return another day.
3. **Language barrier.** Many citizens are most comfortable in Urdu, and many type Urdu in Roman script. Keyword search on websites does not handle spelling variations, synonyms or Roman Urdu.
4. **Load on staff and helplines.** Simple, repetitive enquiries consume the time of counter staff and the helpline.
5. **Risk of misinformation.** General-purpose AI chatbots can give confident but incorrect answers about fees and requirements, because they are not grounded in the official source.

> **Problem in one sentence**
>
> There was no simple, reliable way for a citizen to ask a question about PKM services in their own language and get an accurate answer instantly.

3 Purpose of the Project

The purpose of this project was to **develop an intelligent Urdu assistant that gives citizens accurate, instant answers about Punjab Police Khidmat Markaz services in their own language**, so that they arrive at a centre prepared and spend less time searching for information.

The project also served a technical purpose: to demonstrate that a reliable, low-cost question-answering system for a low-resource language like Urdu can be built by combining multilingual embeddings, rule-based language processing and a large language model grounded in an official knowledge base, without paid infrastructure.

4 Objectives

General objective

To design, develop, test and deploy a conversational Urdu assistant that answers questions about PKM services accurately, using only information from the official PKM website.

Specific objectives

1. **Collect the data**
   Automatically extract service details and centre information from the PKM website using web scraping.
2. **Clean and structure it**
   Convert raw, noisy scraped text into a clean, structured knowledge base with separate fields for documents, fees, time, delivery, addresses and phone numbers.
3. **Understand Urdu questions**
   Handle spelling variations, Arabic-keyboard letters, English loanwords, district aliases and Roman Urdu.
4. **Retrieve relevant facts**
   Find the most relevant information through semantic search with a multilingual sentence-embedding model.
5. **Give focused answers**
   Detect what the user is asking about (fee, documents, time, delivery, age limit, location) and return only that part.
6. **Generate natural answers safely**
   Use a large language model to write fluent Urdu answers strictly from retrieved facts, and refuse out-of-scope questions.
7. **Support conversation**
   Understand follow-up questions using conversation history.
8. **Provide accessible interfaces**
   Build a web application usable on phones and computers, plus desktop and command-line versions.
9. **Measure quality**
   Create an evaluation set of questions to measure accuracy and tune the confidence threshold.
10. **Deploy at no cost**
    Make the system publicly available using free hosting and free-tier services.

5 Scope

In scope

- All 14 services listed on the PKM website
- Required documents, processing fees, processing time, delivery method and age limits
- Address, phone, email and opening hours of 93 centres in 36 districts
- General PKM information: complaints, medico-legal counters, verification and tracking of documents, head office contact
- Questions in Urdu script, and in Roman Urdu or English when the language model is enabled
- Web, desktop and command-line interfaces

Out of scope

- Submitting applications or making payments
- Checking the status of an individual application
- Other police matters such as traffic challans, emergencies or legal advice
- Collecting or storing personal identity information
- Voice input and output
- Real-time synchronisation with the PKM website

6 Target Users and Stakeholders

User / stakeholder	Interest in the system
Citizens of Punjab	Quick answers about services before visiting a centre.
Overseas Pakistanis and their families	Requirements for character certificates and police verification needed for visas and jobs abroad.
Drivers and learners	Licence renewal, duplicate, international and learner licence requirements.
Landlords and employers	Tenant and employee registration procedures.
Khidmat Markaz staff and helpline	Fewer repetitive enquiries.
Developers and researchers	A working reference for Urdu question answering using RAG.

7 Significance and Benefits

Saves time for citizens

Answers in seconds instead of searching through many pages, and fewer wasted trips because people bring the right documents.

Removes the language barrier

Citizens can ask in the language and script they are comfortable with.

Reduces load on staff

Routine questions are answered automatically, so staff can focus on service delivery.

Trustworthy by design

Answers are grounded in official information; the assistant refuses rather than guesses.

Available 24/7

Works outside office hours, on phones and computers.

Zero running cost

Built entirely on free and open-source tools and free tiers.

Contribution to Urdu NLP

Shows practical techniques for handling Urdu spelling variation and Roman Urdu in a real application.

Reusable approach

The same design can serve other government departments with Urdu content.

8 Background Concepts

8.1 Chatbots and question answering

A chatbot is a program that converses with users in natural language. Early chatbots matched keywords against fixed rules; modern systems use machine-learning models that understand meaning. A *closed-domain* question answering system, like this one, answers only within a defined body of knowledge.

8.2 Sentence embeddings and semantic search

A sentence embedding is a list of numbers (a vector) that represents the meaning of a sentence. Sentences with similar meanings have vectors that point in similar directions, so their **cosine similarity** is high. Semantic search compares the question's vector with every document's vector and returns the closest matches, even when they share no exact words. This project uses **paraphrase-multilingual-mpnet-base-v2**, a Sentence-BERT model trained for more than 50 languages including Urdu [2, 3].

8.3 Large language models (LLMs)

LLMs such as Google's Gemini generate fluent text and understand informal language, but they can also produce plausible-sounding false statements, known as *hallucinations*, especially about specific facts such as fees.

8.4 Retrieval-Augmented Generation (RAG)

RAG combines the two approaches [1]. The system first **retrieves** relevant documents from a trusted knowledge base, then passes them to the LLM, which **generates** an answer from those documents only. This keeps answers accurate and up to date without retraining the model, and makes it possible to trace every answer back to its source.

8.5 Challenges of Urdu text

- Right-to-left Perso-Arabic script, with letters that change shape by position.
- Many valid spellings of the same word (سرٹیفکیٹ / سرٹیفیکیٹ) and frequent English loanwords.
- Characters that look identical but have different Unicode codes (Arabic ي vs Urdu ی).
- Optional diacritics, and widespread use of Roman Urdu in digital communication.

9 Methodology

The project was carried out in eight phases, each building on the previous one.

PHASE 1

Requirement analysis

Studied the PKM website, identified the services, the information users need and the common question types.

PHASE 2

Data collection

Built Selenium scrapers to extract the 14 services, 36 district pages and the home page in Urdu.

PHASE 3

Data cleaning

Separated labels from values, corrected spellings, split phone numbers, parsed dates and removed menu noise.

PHASE 4

Knowledge base

Structured the data into services, centres and general facts; created 111 searchable entries.

PHASE 5

Retrieval engine

Implemented normalisation, semantic search, district and aspect detection, and decision rules.

PHASE 6

LLM integration

Added Gemini for question rewriting and grounded answer generation, with automatic fallback.

PHASE 7

Interfaces

Developed the Gradio web app, the Tkinter desktop app and the command-line mode.

PHASE 8

Testing and deployment

Ran functional tests and two rounds of evaluation (26/30, then 33/33) and prepared one-command deployment.

9.1 Data collection

The PKM website loads content dynamically through tabs, so a browser-automation approach was used. Selenium opened each service and district tab in Urdu, and BeautifulSoup extracted headings, lists and paragraphs. The output was saved as raw JSON.

9.2 Data cleaning and structuring

The raw data contained several defects, listed in section 14. A cleaning module converted it into three structured files: `services.json` (14 services, each with sections for documents, fee, processing time, delivery and age limit), `centers.json` (36 districts, 93 centres with address, phone list, email and hours) and `general_info.json` (seven general facts).

9.3 Knowledge representation

Because the embedding model reads only about 128 tokens, each service was represented both as a whole and as separate section entries (e.g. “Character Certificate – Fee: …”). Each entry stores a short *search text* used for matching and a full *answer text* shown to the user. In total, 111 entries were indexed. The embeddings are cached on disk and rebuilt automatically when the data or model changes.

9.4 Question processing and answer selection

Each question is normalised, embedded and compared with all entries by cosine similarity. In parallel, rule-based detectors find district names (with aliases and English names), exact service names, and aspect words such as فیس (fee) or کاغذات (documents). A decision procedure combines these signals to choose a district's centres, a general fact, one section of a service or a full service card. Below a similarity threshold of 0.45, the assistant declines to answer.

9.5 Grounded answer generation

When Gemini is configured, the selected facts, the last three conversation turns and the user's question are sent with instructions to answer in Urdu using only the given information. Roman Urdu and follow-up questions are first rewritten into complete Urdu questions. If the model is unavailable, the retrieved facts are shown directly.

10 System Architecture and Design

The system follows a layered, modular design. Each layer can be changed or tested independently: the interfaces share one core, and the language model is an optional plug-in.

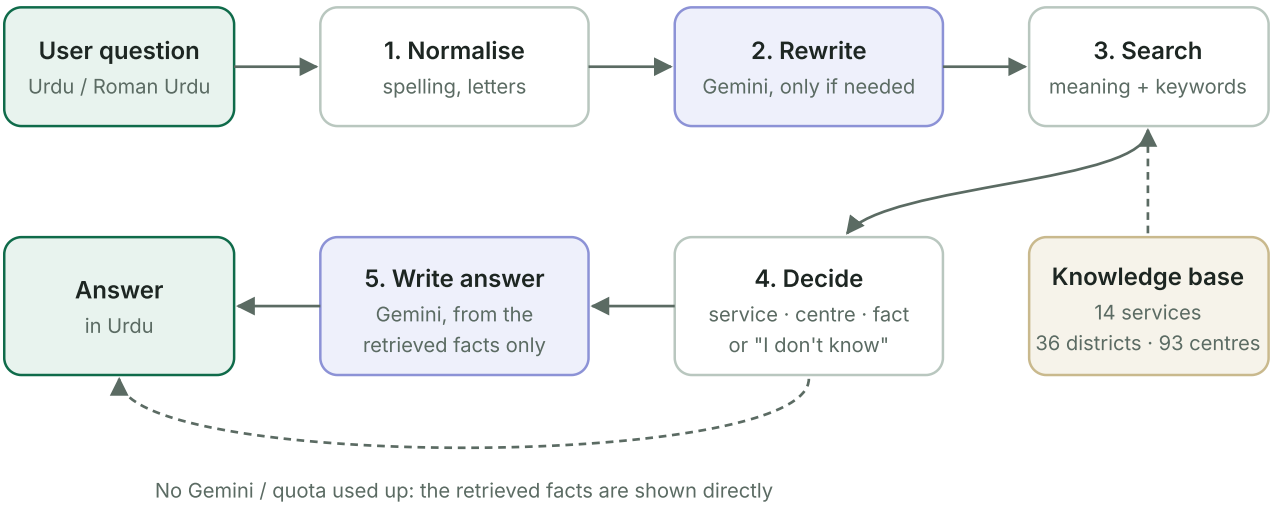


Figure 1. Flow of a question through the system. Purple components use Gemini and are optional.

10.1 Modules

Layer / module	Responsibility
Data acquisition `extract_*.py`	Scrapes services, centres and home page from the PKM website
Data processing `clean_data.py`	Cleans and structures raw data into the knowledge base
Language utilities `text_utils.py`	Spelling correction, normalisation, variant and district alias dictionaries
Core engine `chatbot_core.py`	Indexing, semantic search, intent detection, decision rules, answer formatting, conversation history
LLM layer `llm.py`	Question rewriting and grounded answer generation with Gemini
Presentation `app.py` , `urdu_chatbot.py`	Web, desktop and command-line interfaces
Quality and operations `evaluate.py` , `deploy_hf.py`	Accuracy evaluation and deployment

10.2 Decision logic

Condition	System response
District named and question about location, contact or hours	That district's centres
Centre asked about without a district	Ask the user for the district
Best similarity below threshold	Polite refusal with list of available topics
General topic matched	The general fact
Service and a specific aspect	Only that section of the service
Service without a specific aspect	Full service card

11 Features of the System

Feature	Description
Complete service coverage	All 14 PKM services with documents, fee, time, delivery and age limits.
Centre directory	93 centres in 36 districts with address, phones, email and hours.
Focused answers	Returns only the part asked about (fee, documents, time, etc.).
Robust Urdu understanding	Spelling variants, Arabic-keyboard letters, loanwords and district aliases.
Natural answers	Fluent Urdu responses written by Gemini from official facts only.
Roman Urdu support	“character certificate ki fees kitni hai” is understood.
Follow-up questions	“اس کی فیس؟” resolved from conversation history.
Safe refusals	Questions outside the knowledge base are declined politely.
Graceful fallback	Works without internet or when the LLM quota is exhausted.
Three interfaces	Mobile-friendly web app, desktop app and command line.
User feedback	👍 / 👎 on each answer, logged for improvement.
Fast start-up	Search index cached and rebuilt only when data changes.
One-command deployment	Publishes to Hugging Face Spaces with the API key stored as a secret.

11.1 User interface

The screenshots below were taken from the running web application.



Figure 2. Home screen of the web application: title, description, conversation area, question box, send button, example questions and new-chat button.



Figure 3. Focused answers written by Gemini from the retrieved facts: a fee question returns only the fee, and a documents question only the required documents for licence renewal. Feedback buttons appear below each answer.

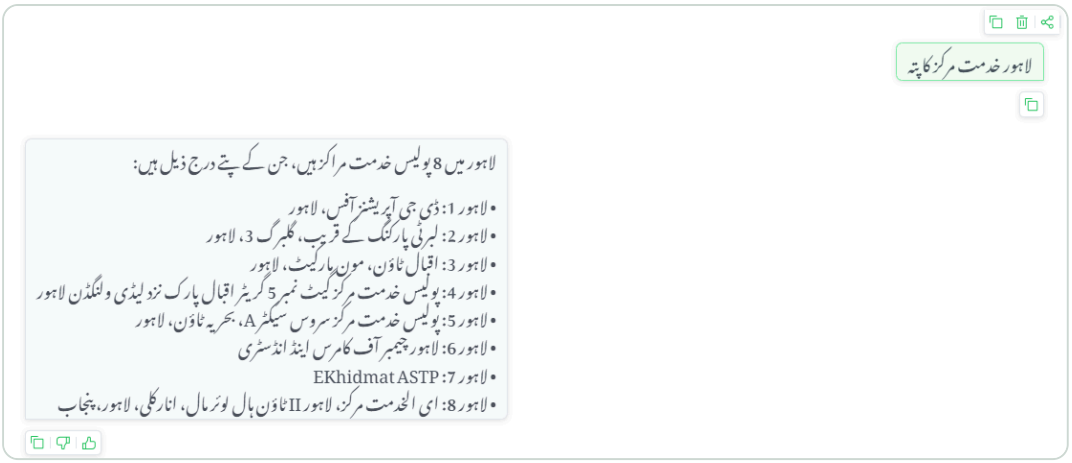


Figure 4. A district question returns all 8 Lahore centres with their addresses.

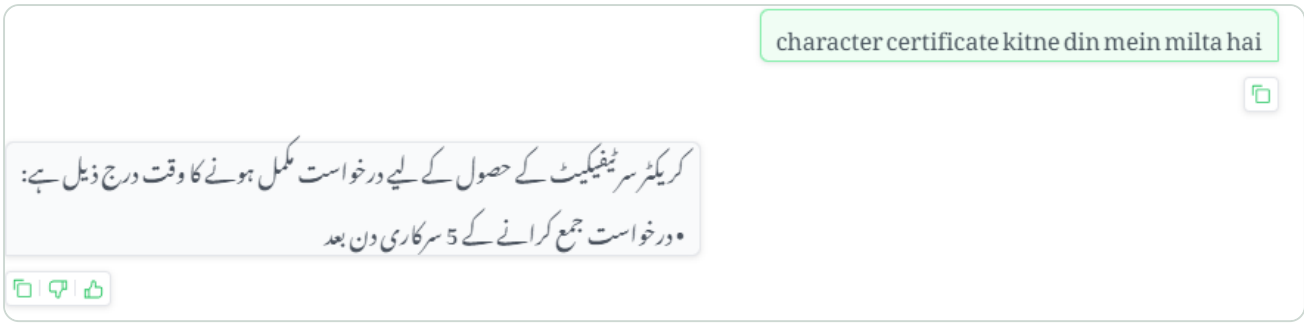


Figure 5. Roman Urdu support: the question “character certificate kitne din mein milta hai” is rewritten into Urdu, matched to the processing time of Character Certificate, and answered in Urdu script.

12 Tools and Technologies

Technology	Category	Use in the project
Python 3.11 – 3.14	Language	All components
Selenium	Web automation	Opening dynamic tabs on the PKM website
BeautifulSoup	HTML parsing	Extracting text from pages
Sentence-Transformers	NLP library	Running the embedding model
paraphrase-multilingual-mpnet-base-v2	Embedding model	Converting Urdu text into meaning vectors
NumPy	Numerical computing	Cosine-similarity search and index cache
Google Gemini (Flash)	Large language model	Question rewriting and answer generation
google-genai SDK	API client	Communicating with Gemini
Gradio 6	Web UI framework	Chat web application
Tkinter	GUI toolkit	Desktop application
Hugging Face Spaces	Cloud hosting	Free public deployment
JSON / JSONL	Data formats	Knowledge base, chat and feedback logs

Hardware and software environment

Development	Windows PC with Python 3.14 (also runs on Linux and macOS)
Deployment	Hugging Face Spaces, CPU basic (2 vCPU, 16 GB RAM), free tier
Client	Any modern web browser on phone or computer

13 Testing and Evaluation

13.1 Testing approach

Testing was carried out at three levels:

- **Unit-level checks** of the cleaning functions: label splitting, phone parsing and date parsing.
- **Functional tests** of the question-routing logic, the LLM integration paths, caching and multi-user behaviour.
- **Accuracy evaluation** with a labelled set of questions (`evaluate.py`), run twice: once on 30 questions, then on 33 questions after fixing the errors found.

13.2 Functional test results

Test	Input	Expected behaviour	Result
Service + aspect	کریکٹر سرٹیفیکیٹ کی فیس کتنی ہے؟	Only the fee section of Character Certificate	**Pass**
Spelling variant	کریکٹر سرٹیفکیٹ کے لیے کاغذات	Variant matched; document list returned	**Pass**
District, Urdu	لاہور خدمت مرکز کا پتہ	All 8 Lahore centres with contacts	**Pass**
District, alias	ڈیرہ غازی خان کا فون نمبر	Mapped to ڈی جی خان centres	**Pass**
District, English	pindi khidmat markaz phone	Mapped to Rawalpindi centres	**Pass**
No district given	خدمت مرکز کہاں ہے	Asks the user for their district	**Pass**
Keyword false match	فیصل آباد میں خدمت مرکز کہاں ہے	“فی” (fee) not triggered inside فیصل	**Pass**
Service + district	لاہور میں کریکٹر سرٹیفیکیٹ کی فیس	Fee + note that any centre can serve	**Pass**
Missing data	ڈرائیونگ لائسنس کی توثیق کی فیس	Full card + note that fee is not listed	**Pass**
Out of scope	آج موسم کیسا ہے	Polite refusal; LLM not called	**Pass**
History	پچھلا سوال کیا تھا	Previous question recalled	**Pass**
Roman Urdu (LLM)	character certificate kitne din mein milta hai	Rewritten to Urdu, processing time answered	**Pass (Fig. 5)**

Follow-up (LLM)	اس کی فیس کتنی ہے؟	Resolved using conversation history	**Pass***
LLM failure	Gemini returns an error	Retrieved facts shown with a notice	**Pass (Fig. 6)**
Index cache	Second start-up	0 documents re-embedded	**Pass**
Multi-user web	Two visitors at once	Separate histories, no mixing	**Pass**

* Tested with a simulated language model. The other Gemini paths were confirmed in real use (Figures 5 and 6).

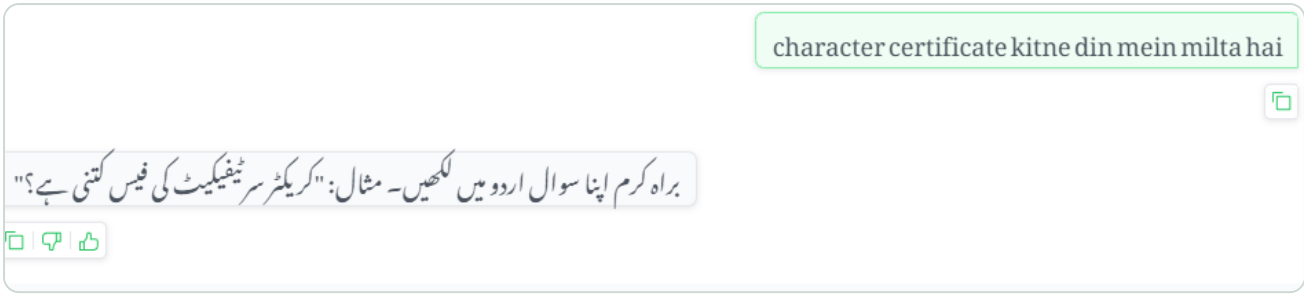


Figure 6. Graceful fallback observed in real use: while Gemini was unavailable, a Roman Urdu question was answered with a polite request to write in Urdu script instead of an error.

13.3 Accuracy evaluation set

The evaluation set contains questions with known correct answers, written the way real users ask, including spelling variations (سرٹیفکیٹ), English loanwords (رینیو, انٹرنیشنل) and district aliases. Each answer counts as correct only if it comes from the expected source: the right service and, where asked, the right section.

Category	Run 1	Run 2	What is checked
Service questions	15	16	Correct service and section (fee, documents, time, age)
Centre questions	6	6	Correct district, including aliases (پنڈی, ڈیرہ غازی خان)
General information	5	7	Complaints, tracking, verification, any-district service, medico-legal
Out-of-scope questions	4	4	Weather, capital city, recipes, cricket: must not be answered
Total	**30**	**33**	

13.4 Evaluation results

86.7%	100.0%	4 → 0	0	33
Run 1 accuracy (26/30)	Run 2 accuracy (33/33)	errors fixed	wrong answers to out-of-scope questions	test questions

Precision, recall and F1 by category

Category	Run 1 (30 questions)				Run 2 (33 questions)			
	Correct	Precision	Recall	F1	Correct	Precision	Recall	F1
Service	14/15	0.78	0.93	0.85	16/16	1.00	1.00	**1.00**
Centre	6/6	1.00	1.00	1.00	6/6	1.00	1.00	**1.00**
General info	2/5	1.00	0.40	0.57	7/7	1.00	1.00	**1.00**
Refused	4/4	1.00	1.00	1.00	4/4	1.00	1.00	**1.00**
Overall accuracy			**26/30 = 86.7%**				**33/33 = 100.0%**	

Precision: of the answers given in a category, how many were correct. Recall: of the questions in a category, how many were answered correctly. F1: their harmonic mean.

Confusion matrix

Run 1: 26 / 30 correct

Expected ↓ / Predicted →	Service	Centre	General info	Refused	Total
Service	15	·	·	·	15
Centre	·	6	·	·	6
General info	3 error	·	2	·	5
Refused	·	·	·	4	4

Run 2: 33 / 33 correct

Expected ↓ / Predicted →	Service	Centre	General info	Refused	Total
Service	16	·	·	·	16
Centre	·	6	·	·	6
General info	·	·	7	·	7
Refused	·	·	·	4	4

Rows show the expected category, columns the category the assistant answered from. The diagonal is correct. Orange-outlined cells are errors. In run 1, three general-information questions were answered from a service, and one service question (in the diagonal) picked the wrong service, so the diagonal alone shows 27 but only 26 were fully correct.

13.5 Error analysis

All four errors in run 1 shared one cause: the question contained a decisive keyword, but the semantic search matched it to a service whose wording was more similar overall.

Question	Expected	Predicted (run 1)	Score	Fix applied
لائسنس گم ہو گیا ہے نقل کیسے بنے گی	نقل ڈرائیونگ لائسنس	ڈرائیونگ لائسنس کی تجدید	0.724	Keyword rule لائسنس + گم / نقل → Duplicate Licence
اپنی درخواست کا اسٹیٹس کیسے چیک کروں	درخواست کی ٹریکنگ	ملازم کی رجسٹریشن	0.708	Keyword rule اسٹیٹس / ٹریک → Application Tracking
کریکٹر سرٹیفیکیٹ اصلی ہے یا نہیں کیسے تصدیق کریں	دستاویز کی تصدیق	کریکٹر سرٹیفیکیٹ	0.631	Keyword rule اصلی / جعلی → Document Verification
میڈیکو لیگل سرٹیفیکیٹ کہاں سے ملے گا	میڈیکو لیگل سرٹیفیکیٹ	کریکٹر سرٹیفیکیٹ	0.737	Keyword rule میڈیکو → Medico-legal

Each error was fixed with a keyword rule that is checked before the semantic search. To make sure the fixes generalise rather than only memorise the failed questions, three new questions with *different wording* were added to the test set:

- ڈپلیکیٹ لائسنس کیسے بنے گا (Duplicate Licence)
- میری درخواست کہاں تک پہنچی ٹریک کیسے کروں (Application Tracking)
- جعلی کریکٹر سرٹیفیکیٹ کی پہچان کیسے کریں (Document Verification)

All three passed in run 2, and none of the 26 questions that passed in run 1 failed.

13.6 Similarity scores and the threshold

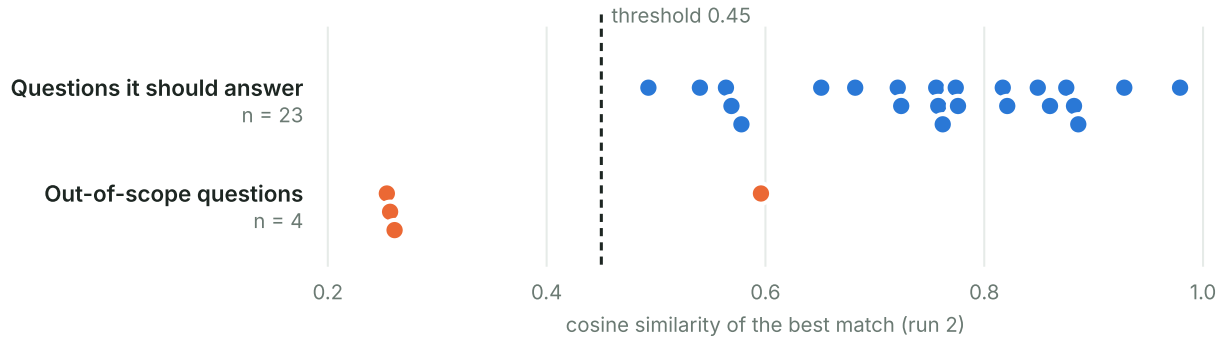


Figure 7. Cosine similarity of the best match for each question in run 2 (questions answered by a keyword rule have no score and are not shown). Blue: questions the assistant should answer; orange: out-of-scope questions.

Valid questions score between 0.49 and 0.98, and three of the four out-of-scope questions score around 0.26, well below the threshold of 0.45. One out-of-scope question, پاکستان کا دارالحکومت کیا ہے (score 0.596), scored above the threshold, but the decision rules still did not return a service answer: the assistant asked the user for a district instead. The threshold alone therefore does not separate every case, and the rule layer acts as a second safeguard. A larger set of out-of-scope questions would allow the threshold to be tuned more precisely.

13.7 Data quality results

Check	Outcome
Services extracted and structured	**14 / 14**
Districts with centre data	**36 / 36**
Centres with separated address, phone and email fields	**93 / 93**
Website spelling errors corrected	**All detected patterns**
Opening dates parsed unambiguously	25 of 36 (11 ambiguous dates intentionally left empty)

14 Challenges and Solutions

Challenge	Solution adopted
Scraped text had labels glued to values and words joined together (e.g. ایڈریسجنرل بس اسٹینڈ).	Rewrote the scrapers to keep separators and spaces, and built a cleaning script that splits labels into structured fields (address, phone, email).
The website itself contains misspellings (ڈرائیونگ, ٹائیم) and wrong district spellings.	Explicit and pattern-based spelling corrections applied during cleaning; district names standardised.
Users write the same word in many ways, use Arabic keyboards, or use English loanwords (“renew”).	A normalisation layer applied to both questions and documents: letter mapping, diacritic removal and a variant dictionary.
The embedding model reads only about 128 tokens, so long documents were cut off.	Knowledge split into short, focused search texts (one per service section), while full cards are kept for answers.
Keyword rules triggered inside other words (the fee word فی inside فیصل آباد).	Whole-word matching with Unicode word boundaries.
Large language models can invent fees or phone numbers (hallucination).	Retrieval-Augmented Generation: the model receives only retrieved facts, with strict instructions, and out-of-scope questions are refused before the model is called.
Roman Urdu and follow-up questions could not be matched by search.	The LLM rewrites such questions into complete Urdu questions before retrieval.
Opening dates had lost their separators (“4112017”) and could be read in two ways.	Dates accepted only when exactly one reading is valid; ambiguous dates are left empty rather than guessed.
Several users chatting at once on the web would share one conversation memory.	A separate history object is kept for each visitor session.

15 Ethical Considerations

- **Accuracy and transparency.** Answers are generated only from official information, and the interface states clearly that the assistant is an independent project and that details should be confirmed with a centre or the helpline 04299211356.
- **No impersonation.** The system does not present itself as an official Punjab Police service.
- **Privacy.** The assistant never asks for personal information. Users are advised not to enter CNIC or passport numbers, since questions may be processed by Google's Gemini service, whose free tier may use prompts to improve its products. Logs are stored only on the machine running the application.
- **Responsible scraping.** Only publicly available information was collected, at a low request rate.
- **Honest uncertainty.** The system declines to answer instead of guessing, and does not fill in missing data (for example ambiguous dates).
- **Security.** API keys are kept out of the code, in a local `.env` file or an encrypted hosting secret.

16 Limitations

- **Information only.** The system cannot submit applications, accept payments or check the status of a specific application.
- **Static knowledge.** Answers are as current as the last data refresh; changes on the website require re-running the scrapers.
- **Limited domain.** Only PKM services and centres are covered.
- **Dependence on an external LLM** for Roman Urdu, follow-ups and natural phrasing; free-tier quotas limit usage, although the system falls back gracefully.
- **Gaps in the source data.** Some details are missing on the website itself (e.g. the fee for driving licence verification) and could not be added.
- **Text only.** Users who cannot read or type are not yet served.
- **Small test set.** The 33-question evaluation set was written by the developer, and the keyword rules were added in response to its errors. Accuracy on real users' questions may be lower and should be measured from the chat and feedback logs.

17 Future Work

WhatsApp integration

Deliver the assistant on WhatsApp, where most citizens already communicate, through the WhatsApp Cloud API.

Voice input and output

Speech recognition and Urdu text-to-speech for users with limited literacy.

Nearest-centre finder

Use map coordinates and the user's location to recommend the closest centre.

Clarifying questions

Ask which licence type is meant when a question is ambiguous, instead of choosing one.

Automatic data refresh

Scheduled scraping with alerts when fees or requirements change.

Hybrid search

Add keyword search (BM25) alongside semantic search for exact terms such as “FIR” or “LTV”.

Wider coverage

Traffic police licensing, complaint systems and other departments' Urdu services.

Local LLM fallback

A locally hosted open model for when the cloud quota is exhausted.

18 Conclusion

The PKM Urdu Assistant was successfully developed and makes Punjab Police Khidmat Markaz information accessible through natural conversation in Urdu. Starting from information scattered across the PKM website, the project built a clean, structured knowledge base of 14 services and 93 centres, and a question-answering engine that combines multilingual semantic search with Urdu-specific language processing.

By applying Retrieval-Augmented Generation, the system gains the fluency of a large language model while keeping its answers grounded in official facts, and it declines questions outside its knowledge rather than guessing. Spelling variations, Roman Urdu and follow-up questions, which are common obstacles for Urdu applications, are handled through normalisation and query rewriting.

The assistant is available as a mobile-friendly web application, a desktop application and a command-line tool, and can be deployed at no cost. Its evaluation improved from 26/30 to 33/33 correct answers after one round of error analysis. Beyond its practical value to citizens, the project shows that reliable AI assistants for Urdu-speaking users can be built with free and open-source tools, and it provides a foundation that can be extended to WhatsApp, voice interaction and other public services.

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